ASSIGNMENT REPORT-3

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For The Subject

Pattern Recognition and Machine Learning

(26AI5707)

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INTRODUCTION:

[**WRITE HERE:** *Briefly introduce polynomial regression and state the objective of the assignment.*]

Experiments Performed:

Dataset and Initial Observation

[**WRITE HERE:** *Describe the assigned dataset, its two columns, and what the scatter plot suggests about the relationship between x and y.*]

The assigned file contains $N = 10{,}000$ input-target pairs:

$$\mathcal{D} = \{(x_i, y_i)\}_{i=1}^{N}, \qquad N = 10{,}000. \tag{1}$$

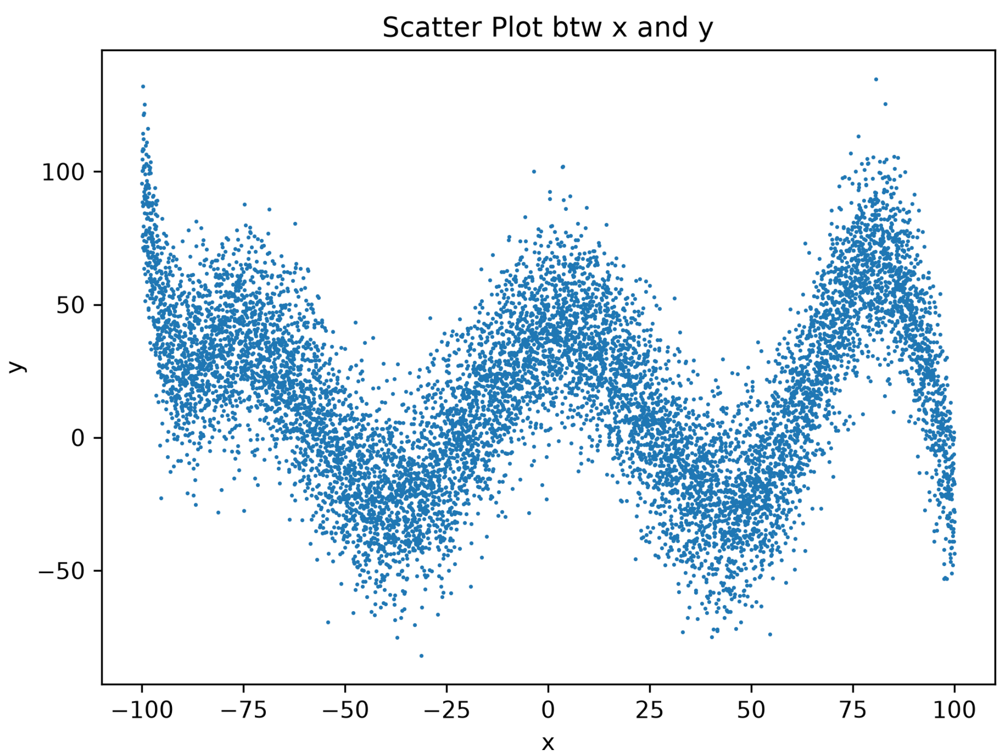

Figure 1: Scatter plot of the input x and noisy target y from `noisy_2.txt`.

Data Shuffling and Splitting

[**WRITE HERE:** *Explain why the data was shuffled and record the train, validation, and test split used in the experiment.*]

With a fixed random seed of 42, the shuffled dataset was divided as follows:

Split	Fraction	Number of samples
Training	60%	6,000
Validation	20%	2,000
Test	20%	2,000

Table 1: Train, validation, and test split used in the experiment.

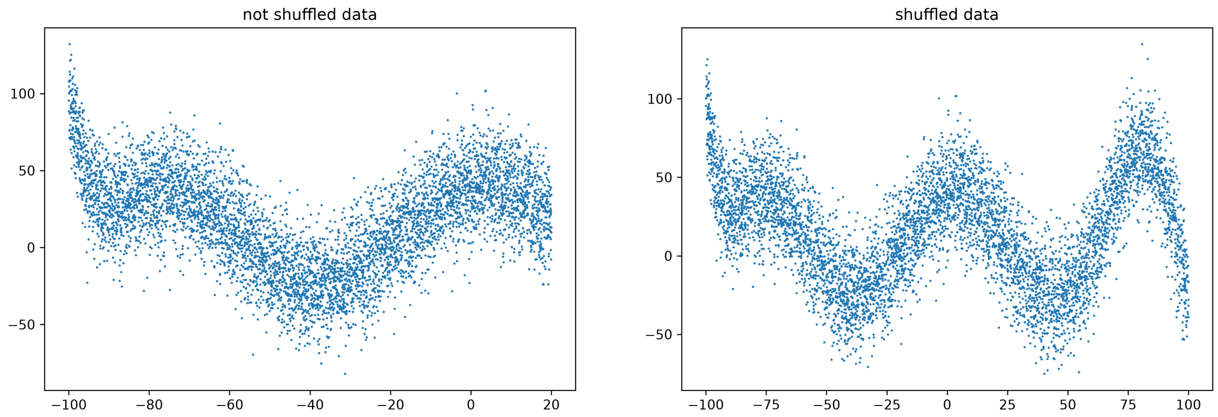

Figure 2: Comparison of the training portion before and after shuffling the dataset.

Polynomial Regression

[**WRITE HERE:** *Explain in your own words why feature scaling, a polynomial basis, and least squares are used here. Interpret the equations below.*]

Feature standardization. Only the training split is used to calculate the mean and standard deviation:

$$\mu_x = \frac{1}{N_{\text{tr}}} \sum_{i=1}^{N_{\text{tr}}} x_i, \qquad \sigma_x = \sqrt{\frac{1}{N_{\text{tr}}} \sum_{i=1}^{N_{\text{tr}}} (x_i - \mu_x)^2}, \tag{2}$$

$$z_i = \frac{x_i - \mu_x}{\sigma_x}, \qquad N_{\text{tr}} = 6{,}000. \tag{3}$$

The same μ_x and σ_x are used to standardize the validation and test inputs.

Polynomial basis. For degree d, each standardized input is mapped to

$$\boldsymbol{\phi}_d(z_i) = \begin{bmatrix} 1 & z_i & z_i^2 & \cdots & z_i^d \end{bmatrix}^T. \tag{4}$$

The design matrix is

$$X_d = \begin{bmatrix} 1 & z_1 & z_1^2 & \cdots & z_1^d \\ 1 & z_2 & z_2^2 & \cdots & z_2^d \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ 1 & z_N & z_N^2 & \cdots & z_N^d \end{bmatrix}. \tag{5}$$

Model and prediction.

$$\hat{y}_i = \boldsymbol{\phi}_d(z_i)^T \mathbf{w} = \sum_{j=0}^{d} w_j z_i^j. \tag{6}$$

Least-squares solution. The weights minimize the mean squared training residual:

$$J(\mathbf{w}) = \frac{1}{N_{\text{tr}}} \|\mathbf{y} - X_d \mathbf{w}\|_2^2. \tag{7}$$

Setting the gradient to zero gives the normal equation and its solution:

$$X_d^T X_d \hat{\mathbf{w}} = X_d^T \mathbf{y}, \tag{8}$$

$$\hat{\mathbf{w}} = (X_d^T X_d)^{-1} X_d^T \mathbf{y}. \tag{9}$$

Mean squared error. For any split $\mathcal{S}$,

$$\text{MSE}_{\mathcal{S}}(d) = \frac{1}{|\mathcal{S}|} \sum_{i \in \mathcal{S}} (y_i - \hat{y}_i)^2. \tag{10}$$

Effect of Polynomial Degree

[**WRITE HERE:** *Discuss underfitting and how the fitted curve changes as the polynomial degree increases.*]

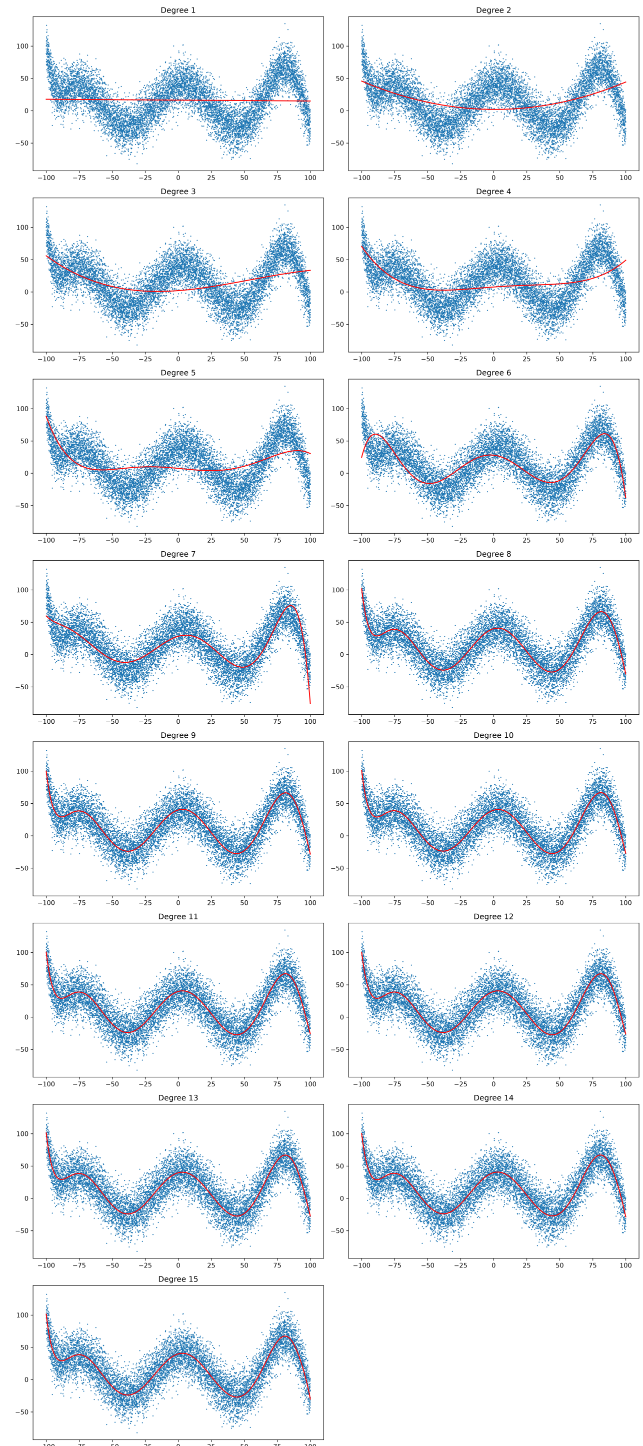

Figure 3: Polynomial curves fitted for degrees 1 through 15.

The degree is selected using the validation error:

$$d^* = \underset{d \in \{1,2,\ldots,15\}}{\arg\min} \; \mathrm{MSE}_{\mathrm{val}}(d) = 10. \tag{11}$$

Degree d	Training MSE	Validation MSE
1	1144.10	1055.51
3	961.71	901.58
5	906.23	849.98
6	567.61	534.78
7	478.17	467.47
8	364.92	348.96
10	364.69	348.63
15	364.53	349.14

Table 2: Selected training and validation errors from the degree experiment.

[**WRITE HERE:** *Compare the training and validation errors, explain the change around degrees 6–8, and justify the selected degree in your own words.*]

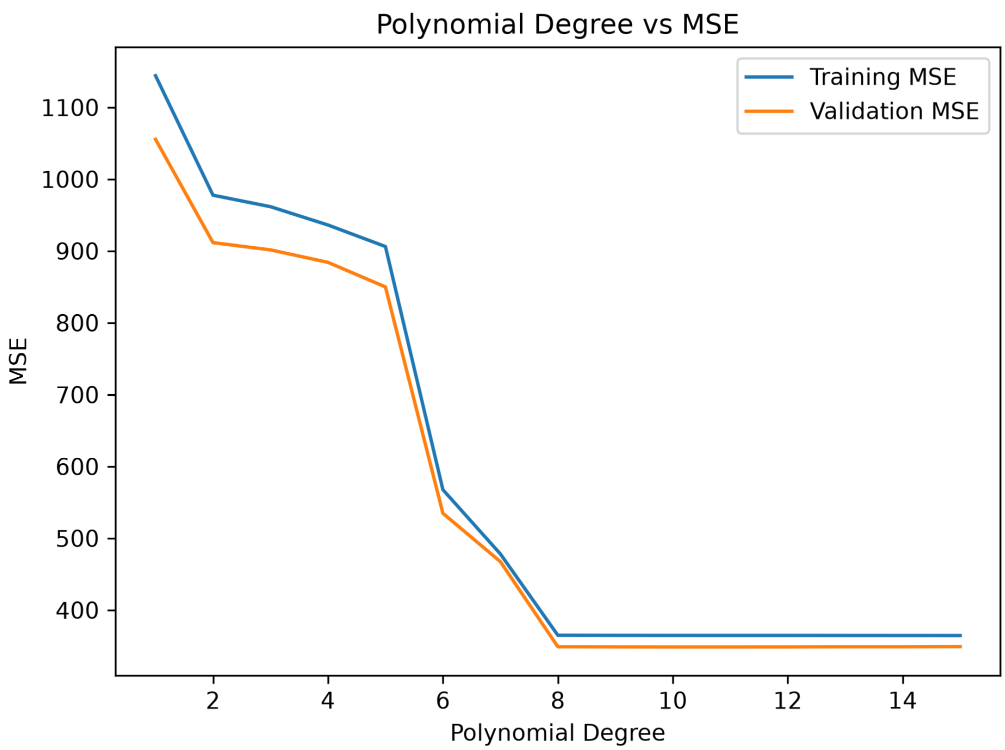

Figure 4: Training and validation mean squared error across polynomial degrees.

Final Model Evaluation

After selecting $d^* = 10$, the training and validation samples are combined:

$$\mathcal{D}_{\text{fit}} = \mathcal{D}_{\text{train}} \cup \mathcal{D}_{\text{val}}, \qquad |\mathcal{D}_{\text{fit}}| = 8{,}000. \tag{12}$$

The standardization parameters and degree-10 weights are recalculated using only $\mathcal{D}_{\text{fit}}$. The resulting held-out test error is

$$\text{MSE}_{\text{test}}(10) = 340.0071. \tag{13}$$

[**WRITE HERE:** *Explain the final fitting procedure and interpret the held-out test result in your own words.*]

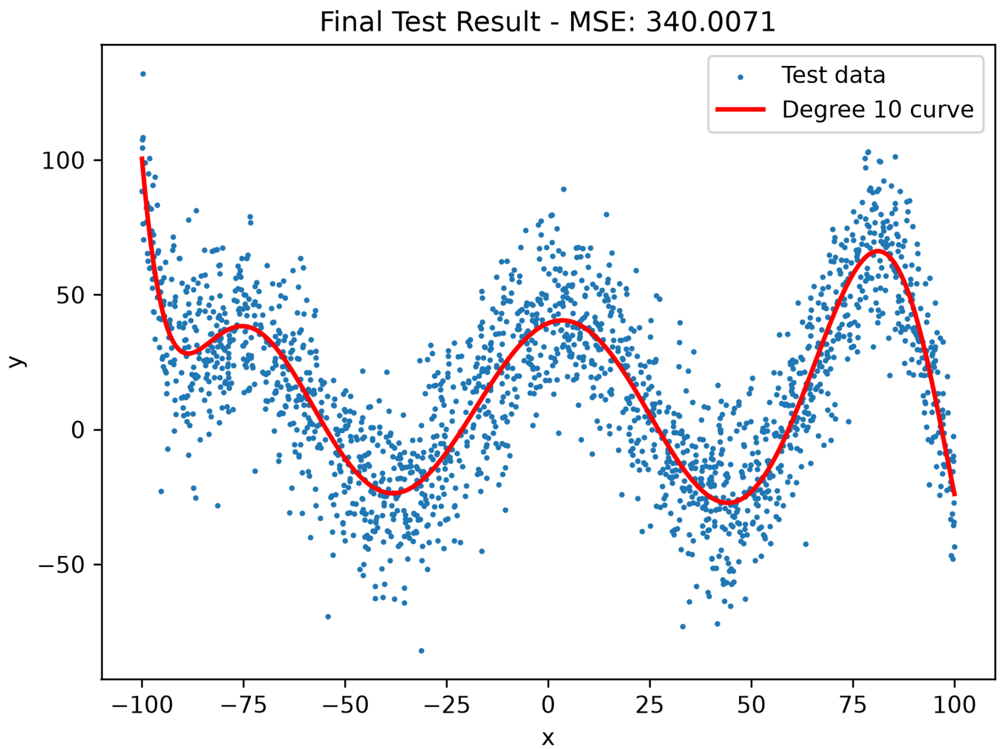

Figure 5: Degree-10 polynomial fitted to the held-out test data; test MSE = 340.0071.

Additional Experiments:

[**WRITE HERE:** *Add any experiments performed beyond the assignment requirements, such as other basis functions, regularization, or different data splits. Delete this section if it is not used.*]

Inference

[**WRITE HERE:** *Summarize your observations about model complexity, the selected degree, generalization, and what you learned from the experiment.*]